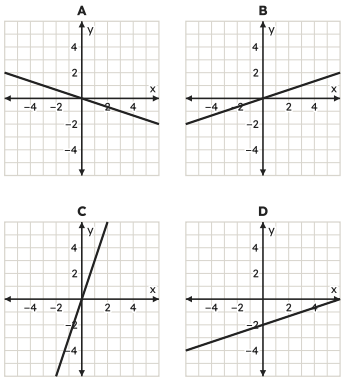
Benchmark: **MA.7.AR.4.3** · Graph proportional relationships Student code: _____ Calculator: yes

Learning goal: I can graph proportional relationships. (circle your level after scoring)

4	I can use a graph to find the constant of proportionality and interpret it in context.
3	I can graph a proportional relationship and identify the points (0, 0) and (1, r).
2	I can plot points from a table.
1	I need help to start this benchmark.

1. A proportional relationship between x and y has a constant of proportionality of $\frac{1}{3}$.

Which graph represents this relationship?



Choose the graph that matches.

A. Graph A

B. Graph B

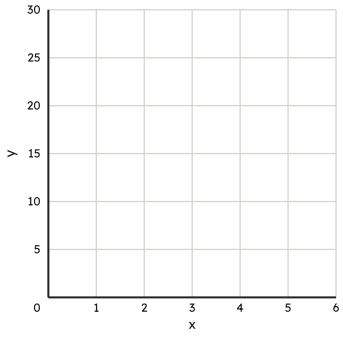
C. Graph C

D. Graph D

2. The table shows a proportional relationship between x and y.

x	y
4	20
5	25
6	30

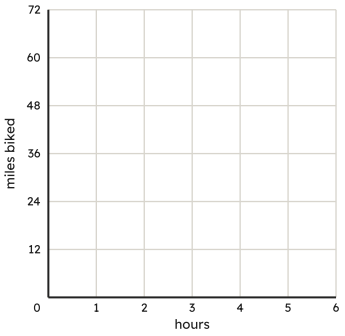
Graph the relationship on the grid.



Turn over →

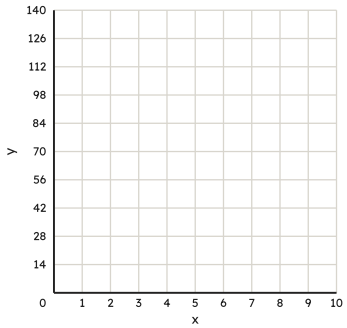
3. The relationship between hours (x) and miles biked (y) is proportional, with a constant of proportionality of 12.

Graph the relationship on the grid for 0 to 6 hours. What does the constant of proportionality mean in this situation?



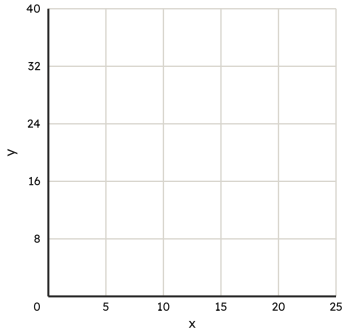
4. A proportional relationship is given by the equation $y = 14x$.

Graph the relationship on the grid for x-values from 0 to 10.



5. The constant of proportionality for a proportional relationship between x and y is $\frac{8}{5}$.

Graph the relationship on the grid for x-values from 0 to 25, and mark and label the point on the line whose x-coordinate is 5. Is the point $(40, 64)$ on the line? Explain.



Score: ____ / 5 My level: 1 2 3 4 → copy both to your progress strip

Answer key — C-AR.4.3 · MA.7.AR.4.3 · Graph proportional relationships

Unit 11: Representing Proportional Relationships. Regular = on-benchmark item; Rigor = the item that needs the benchmark's reasoning. Mastery on the weekly check = 4 of 5.

Weekly check (5 items)

1. Regular **Graph B**

A proportional relationship is a straight line through the origin, so the graph passes through $(0, 0)$.
The constant of proportionality $\frac{1}{3}$ is the slope: from $(0, 0)$ the line rises 1 for every 3 to the right, so it passes through $(3, 1)$, inside the labeled part of each graph.
The reciprocal 3 would pass through $(1, 3)$ instead, which switches x and y; a line that falls to the right has a negative constant.

2. Regular **The points $(4, 20)$, $(5, 25)$, $(6, 30)$ and a straight line through them and the origin.**

Plot each row of the table as a point: $(4, 20)$, $(5, 25)$, $(6, 30)$.
Each y-value is 5 times its x-value, so the line is proportional and passes through $(0, 0)$; draw the straight line through the origin and the points.

3. Regular **A straight line from the origin to $(6, 72)$, through $(1, 12)$, $(2, 24)$ and $(3, 36)$; the constant means each hour, 12 more miles are biked. Score: 1 point for the graph and the meaning of the constant; the graph alone earns 0.**

The constant of proportionality 12 means $y = 12x$: each hour, 12 more miles are biked.
Plot $(0, 0)$, $(1, 12)$, $(2, 24)$ and draw the line through them to $x = 6$.

4. Regular **A straight line from $(0, 0)$ to $(10, 140)$, through points such as $(2, 28)$, $(3, 42)$, $(7, 98)$.**

In $y = 14x$ the constant of proportionality is 14, so every point has $y = 14 \times x$.
Plot $(0, 0)$, $(2, 28)$, $(3, 42)$, $(7, 98)$ and draw the straight line through them from $x = 0$ to $x = 10$.

5. Rigor **A straight line through the origin and $(5, 8)$, $(10, 16)$, $(15, 24)$, continued to $(25, 40)$; the point $(5, 8)$ marked and labeled; yes — $\frac{8}{5} = 1\frac{3}{5} \times 40$ is 64, so $(40, 64)$ is on the line. Score: 1 point for the graph with $(5, 8)$ labeled and the correct yes or no with its reason; the graph alone earns 0.**

The constant of proportionality $\frac{8}{5}$ means $y = \left(\frac{8}{5}\right) x$. Choose x-values that are multiples of 5 so y is a whole number: $x = 5$ gives $y = \frac{8}{5} \times 5 = 8$; $x = 10$ gives $y = \frac{8}{5} \times 10 = 16$; $x = 15$ gives $y = \frac{8}{5} \times 15 = 24$.
Plot those points and $(0, 0)$, then draw the line through them to $x = 25$.
At $x = 5$, $y = \frac{8}{5} \times 5 = 8$, so the point $(5, 8)$ is on the line, where the gridlines $x = 5$ and $y = 8$ cross; mark and label it.
A point is on the line when $y = \frac{8}{5} \times x$: $\frac{8}{5} \times 40 = 64$, which is 64, so $(40, 64)$ is on the line.

Re-check strip (3 items, C-AR.4.3-R)

1. Regular **The points $(2, 32)$, $(4, 64)$, $(5, 80)$ and a straight line through them and the origin.**

Plot each row of the table as a point: $(2, 32)$, $(4, 64)$, $(5, 80)$.
Each y-value is 16 times its x-value, so the line is proportional and passes through $(0, 0)$; draw the straight line through the origin and the points.

2. Regular **A straight line from the origin to $(6, 90)$, through $(1, 15)$, $(2, 30)$ and $(3, 45)$; the constant means each box holds 15 books. Score: 1 point for the graph and the meaning of the constant; the graph alone earns 0.**

The constant of proportionality 15 means $y = 15x$: each box holds 15 books.
Plot $(0, 0)$, $(1, 15)$, $(2, 30)$ and draw the line through them to $x = 6$.

3. Regular **A straight line from $(0, 0)$ to $(6, 138)$, through points such as $(1, 23)$, $(2, 46)$, $(6, 138)$.**

In $y = 23x$ the constant of proportionality is 23, so every point has $y = 23 \times x$.
Plot $(0, 0)$, $(1, 23)$, $(2, 46)$, $(6, 138)$ and draw the straight line through them from $x = 0$ to $x = 6$.

Comparison strip (2 items, C-AR.4.3-X) — used on a later week's re-check when this benchmark was *not* retaught

1. Regular **Graph C**

Each y-value is $\frac{1}{3}$ times its x-value, so the constant of proportionality is $\frac{1}{3}$.
The graph is a straight line through the origin that rises 1 for every 3 to the right, so it passes through $(3, 1)$, inside the labeled part of each graph.
A line through $(1, 3)$ would switch x and y; a line that falls to the right has a negative constant.

2. Regular **Graph A**

A proportional relationship is a straight line through the origin, so the graph passes through $(0, 0)$.
The constant of proportionality 2 is the slope: from $(0, 0)$ the line rises 2 for every 1 to the right, so it passes through $(1, 2)$, inside the labeled part of each graph.
The reciprocal $\frac{1}{2}$ would pass through $(2, 1)$ instead, which switches x and y; a line that falls to the right has a negative constant.

Reteach plan — C-AR.4.3

MA.7.AR.4.3 · Graph proportional relationships

Item that gave the most trouble (from both tally sheets):

What the mistake looks like:

Worked example for Tuesday (8–10 min):

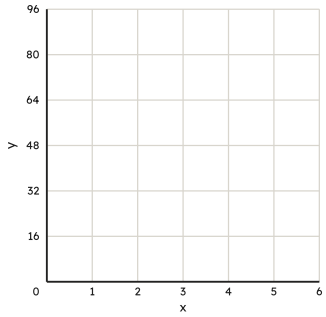
Three practice items (one change at a time):

Decision line: Reteach on _____ Huddle: in person / paper Date: _____ Re-check given on _____

1. The table shows a proportional relationship between x and y.

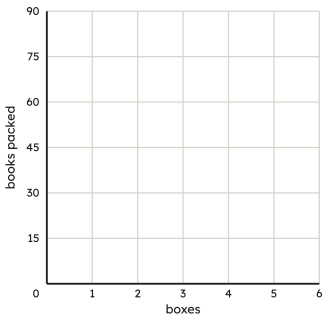
x	y
2	32
4	64
5	80

Graph the relationship on the grid.



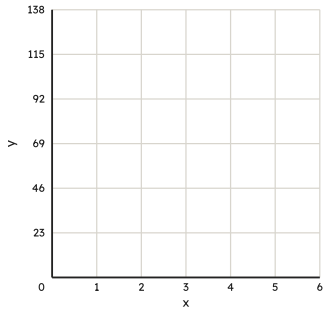
2. The relationship between boxes (x) and books packed (y) is proportional, with a constant of proportionality of 15.

Graph the relationship on the grid for 0 to 6 boxes. What does the constant of proportionality mean in this situation?



3. A proportional relationship is given by the equation $y = 23x$.

Graph the relationship on the grid for x-values from 0 to 6.

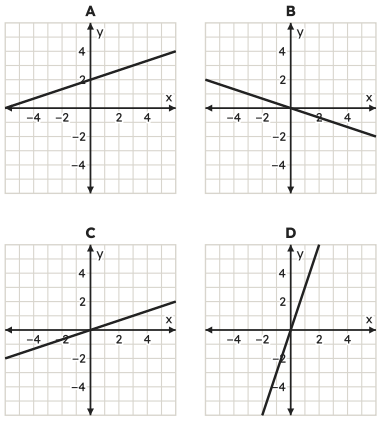


Score: ____ / 3 **→ copy to your progress strip**

1. The table shows a proportional relationship between x and y.

x	y
3	1
9	3
12	4

Which graph represents this relationship?



Choose the graph that matches.

A. Graph A

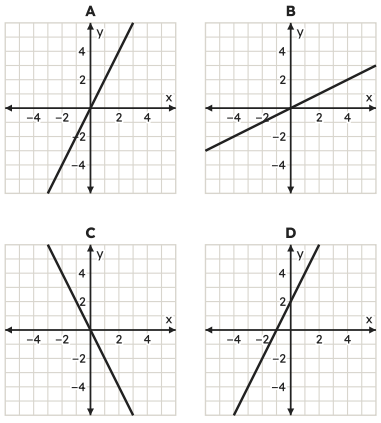
B. Graph B

C. Graph C

D. Graph D

2. A proportional relationship between x and y has a constant of proportionality of 2.

Which graph represents this relationship?



Choose the graph that matches.

A. Graph A

B. Graph B

C. Graph C

D. Graph D

Score: ____ / 2 → **copy to your progress strip**